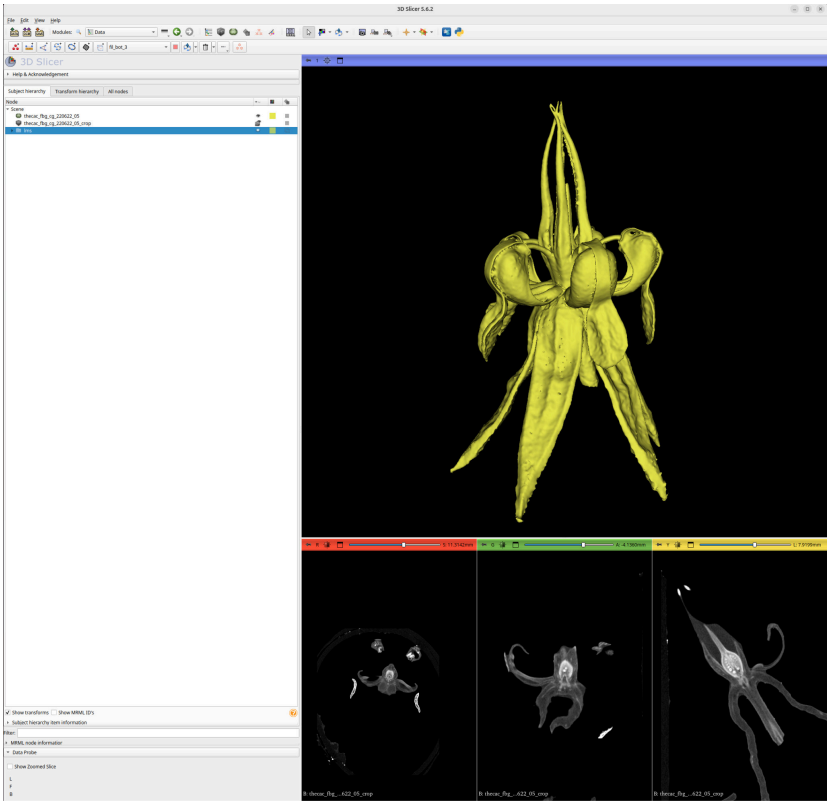
Landmarking flower models in 3D slicer - *Theobroma cacao* - THECAC
Last edited 27 February 2025 by K. Wolcott

Recommended directory structure

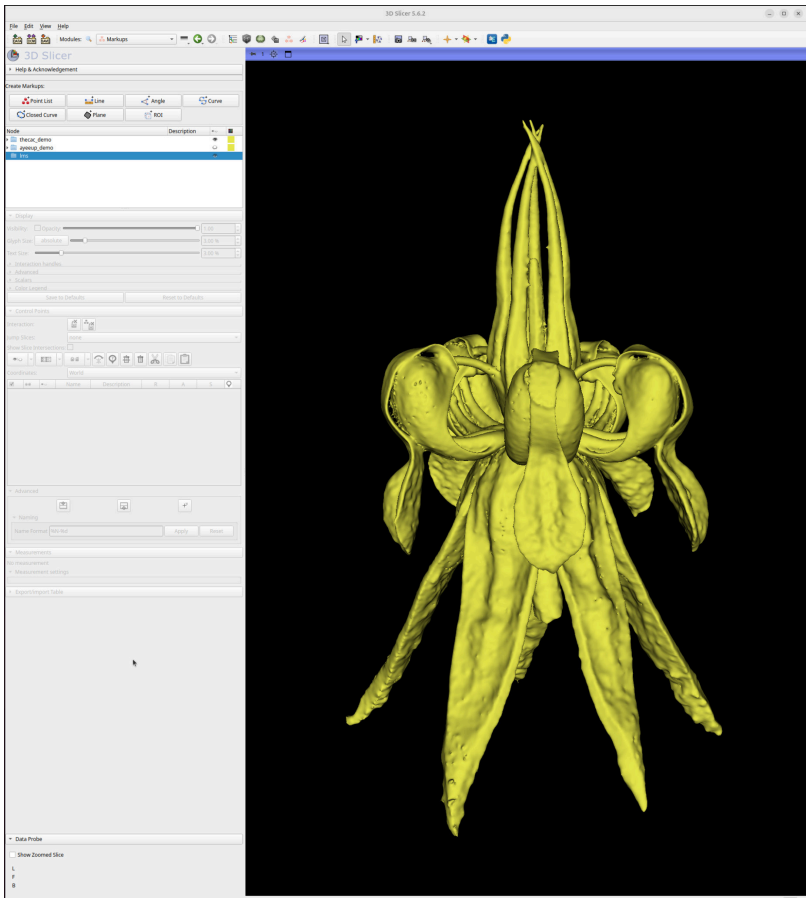
models_for_gmm/
- whole_flower/
  - thecac/
    - Thecac_fbg_cg_220622_05.ply
    - [all models (.ply files) before landmarking]
    - already_lmed/
      - [all models (.ply files) used in analysis, moved here after landmarking is finished]
    - lms/
      - thecac_fbg_cg_220622_05/
        - lms/
          - fil_bot.mrk.json
          - fil_top.mrk.json
          - [all remaining landmark files (.mrk.json)]
      - [all folders for datasets used in analysis with subfolder “lms” containing landmarking files (mrk.json)]
    - exclude/
      - [any models (.ply files) without 1 whorl of floral symmetry to exclude from analysis]

Useful videos: https://www.youtube.com/watch?v=aazPHFZWMdA , https://www.youtube.com/watch?v=xrly9atx7xQ

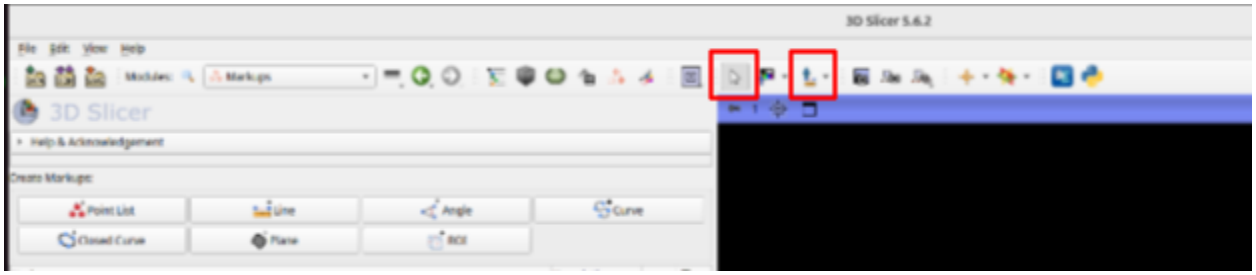
1. **Open 3D Slicer**
2. **Import 3D model of interest & corresponding 3D volume (_cropped.nrrd file)**



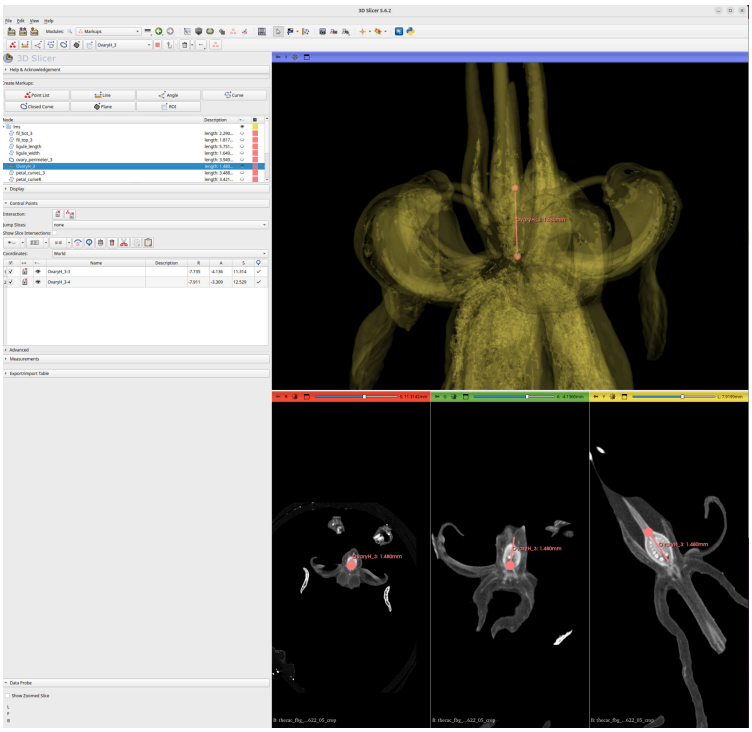
3. **<u>Go to the “Markups” Module</u>**
    a. Inspect the flower in 3D views and confirm that there is at least one whorl with one complete unit of floral symmetry intact (one adjacent petal, sepal, ovary, and staminode)
        i. If not, move the flower model file into the folder, “models_for_gmm/exclude” and begin from Step 1
        ii. If so, randomly select one unit of floral symmetry for applying landmarks
    b. Make a new folder under “Node” -> “lms”



    c. Start placing landmarks
        i. Click on the “Line” tool under “Create Markups”
            1. Rename it “OvaryH” (ovary height)
                a. Click on OvaryH to make sure you add lms to this unit
                b. Reposition the flower looking down over your chosen unit of floral symmetry and zoom into the ovary
                    i. Toggle between the arrow cursor and the landmarking tool in the top menu bar



c. Place one landmark centered at the base of the ovary
d. Place on landmark at the top of the ovary (where the stigma starts tapering)
e. Place one landmark centered at the base of the ovary
    i. Visualize 2D slices until you have nice longitudinal and cross sections of the ovary
    ii. Show the 3D surface model as semi-transparent
        1. Models module
        2. Visibility -> Opacity
            a. Adjust until you can see the ovary
        3. Return to the Markups Module to apply lms
            a. Place landmark at base of ovary in a longitudinal slice view
f. Place one landmark centered at the top of the ovary (before the stigma starts tapering)
    i. Place the landmark on a longitudinal slice view
    ii. Adjust positions of ovary top and bottom landmarks through 2D slice views and 3D view
    iii. In 2D slices, you can click the landmark points and the view in all slices will show where the point is positioned

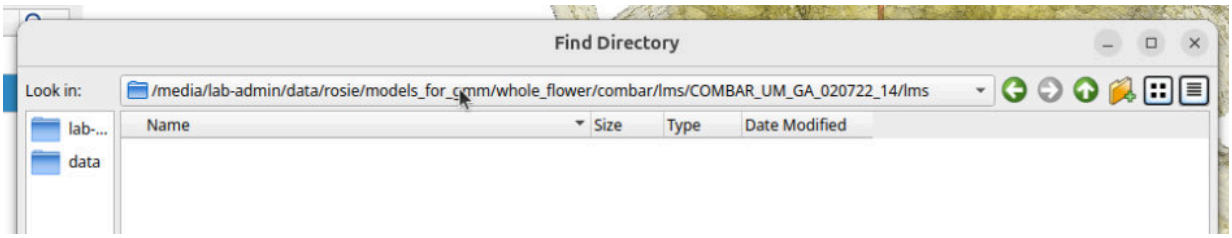


a. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

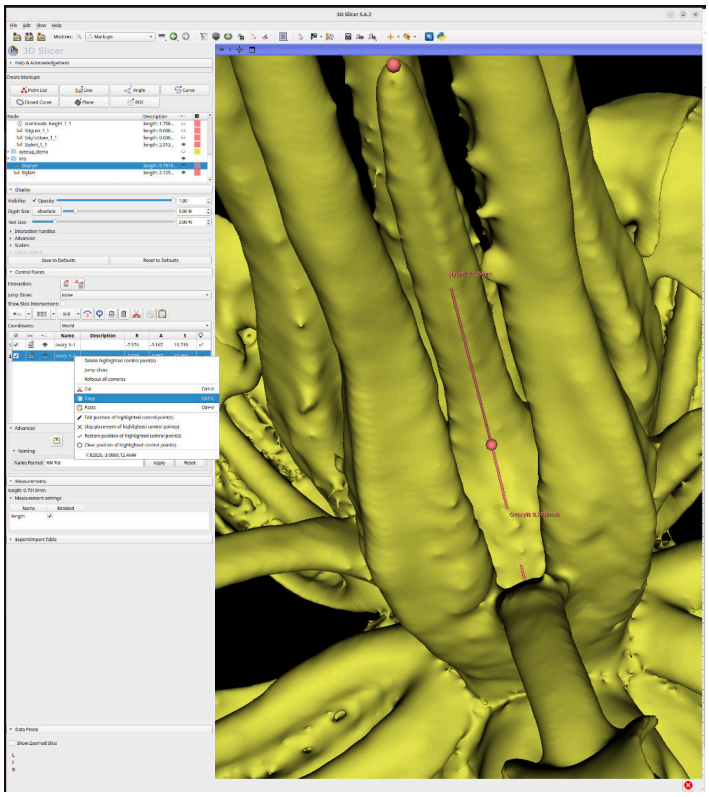
2. Save your results

a. Save -> Change Directory for Selected Files -> Go to models_for_gmm/whole_flower/thecac/lms -> Create a new folder for storing landmarks for this specimen
    i. Name the folder based on the name of your .ply file
        1. Ex: Thecac_fbg_cg_220622_05.ply will have a folder named Thecac_fbg_cg_220622_05
    ii. Create a folder within your dataset folder called "lms"
        1. This is where you will save your landmarks

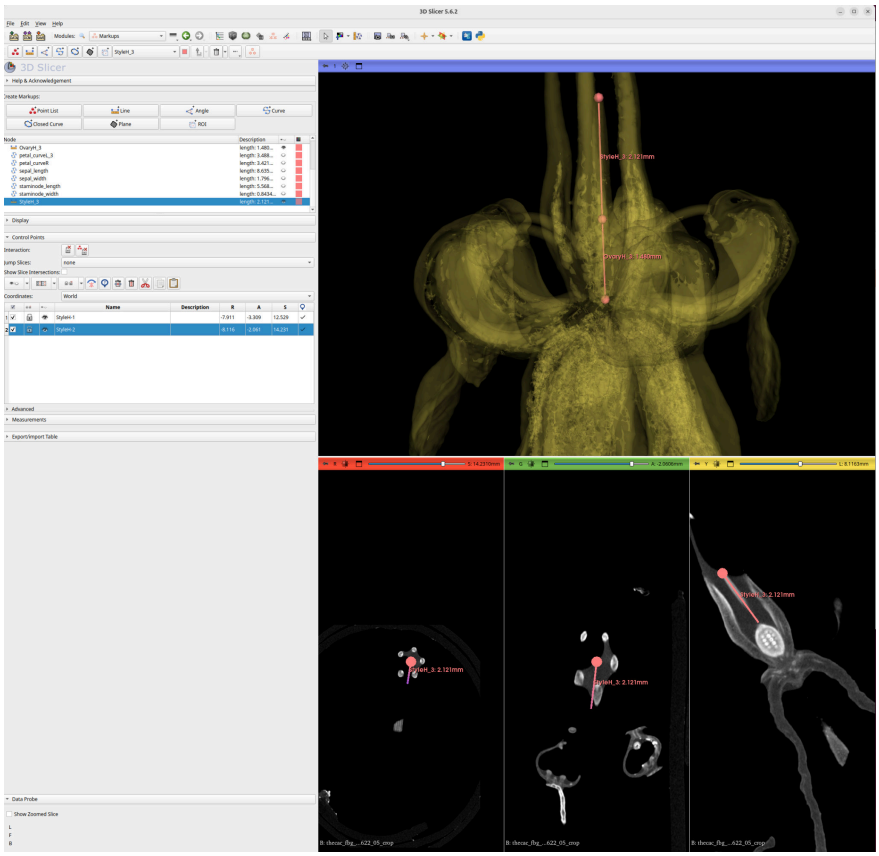
3. **Remember to save results periodically while landmarking!**
    a. **Check that landmarks are being saved to the correct folder for your specimen**
    b. **models_for_gmm/whole_flower/thecac/lms/[specimen_name]/lms/**



4. **Click on the "Line" tool under "Create Markups"**
    a. **Rename it "StyleH" (Style Height)**
        i. Click on StyleH to make sure you are adding lms to this unit
        ii. Add a landmark to the base of the style
            1. Click on OvaryH
            2. Copy the point from the top of the Ovary (ovary_h-2, in this case)

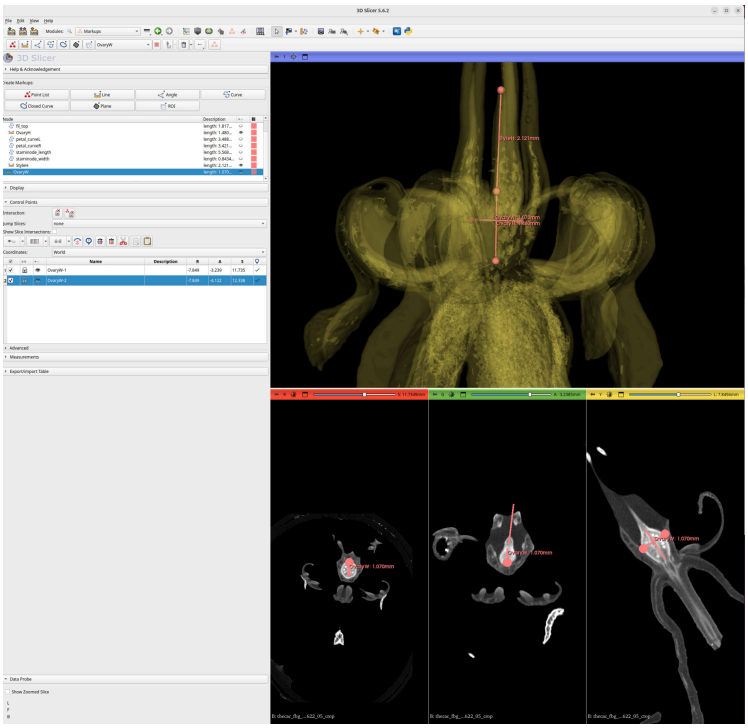


iii. Click on StyleH
    1. Paste the point from the top of the ovary here to be the base of the Style
    2. Rename it to match with the StyleH naming scheme (in this case, StyleH-1)
        a. Click the lock icon next to StyleH-1 to freeze this point
    3. Add another landmark to the top of the style



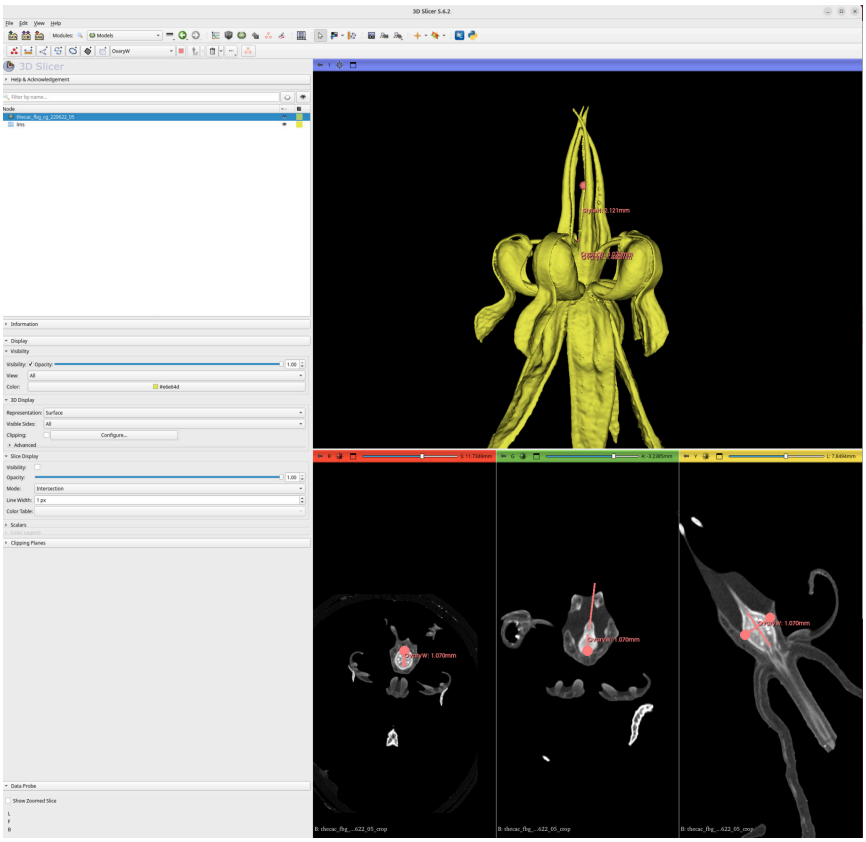
4. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

5. **Click on the “Line” tool under “Create Markups”**
    a. **Rename it to “OvaryW” (Ovary Width)**
        i. Click on OvaryW to make sure you are adding lms to this unit
        ii. Identify the widest part of the ovary
        iii. Add a landmark on each side that are perpendicular to the line for OvaryH
            1. Use longitudinal, cross sections and the semi-transparent surface model for placement

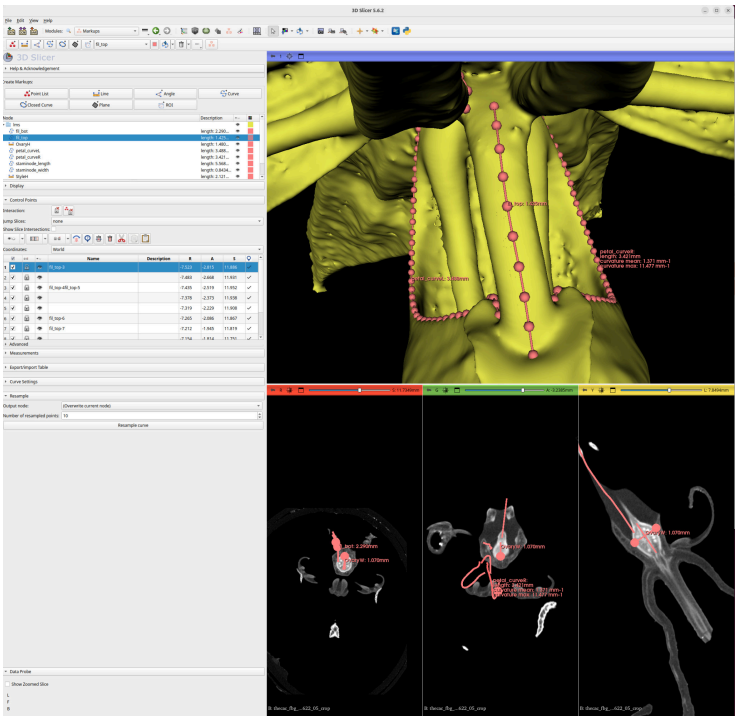


iv. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model
    1. Make sure all landmark points have the closed lock icon next to them

6. **Return to fully opaque surface model view**
    a. Go to “Models” module
        i. Adjust Opacity -> 1.00



7. **Return to the “Markups” Module**
8. **Click on the “Curve” tool under “Create Markups”**
    a. **Rename it to “fil_top” (Filament top)**
        i. Click on “fil_top”
            1. Copy the first point from “ovary_perimeter” to “fil_top”
            2. Rename it to “fil_top-1”
            3. Position the flower with a good view of the top of the filament for landmarking
            4. Add landmarks along the top of the filament from the point where it is detached/free
                a. Follow the midline all the way to the top of the anther



5. Scroll down to the “Resample” tab
    a. Select “fil_top” as “Output Node”
    b. Set the “Number of resampled points” to 10
    c. Click “Resample curve”
    d. Now all points are evenly spaced along the curve
6. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model
    a. Make sure all landmark points have the closed lock icon next to them

9. Click on the “Curve” tool under “Create Markups”

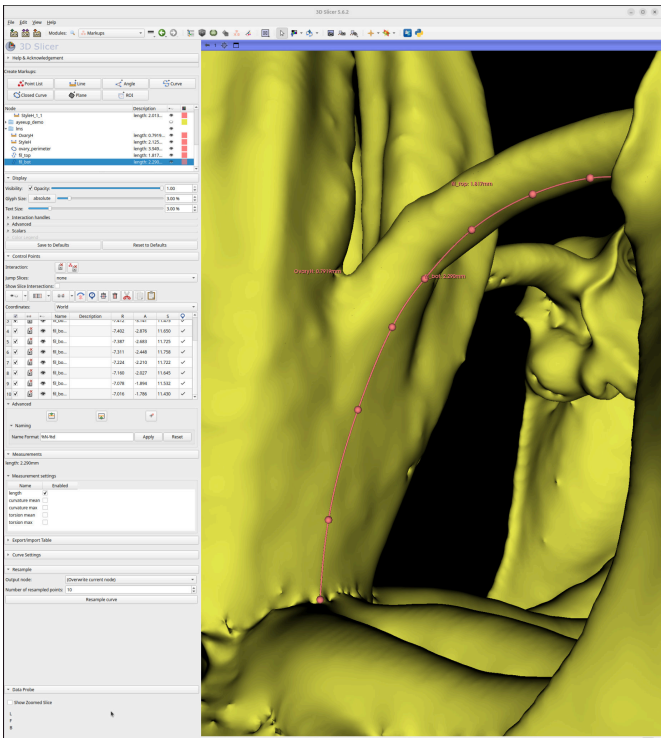
a. Rename it to “fil_bot” (Filament bottom)

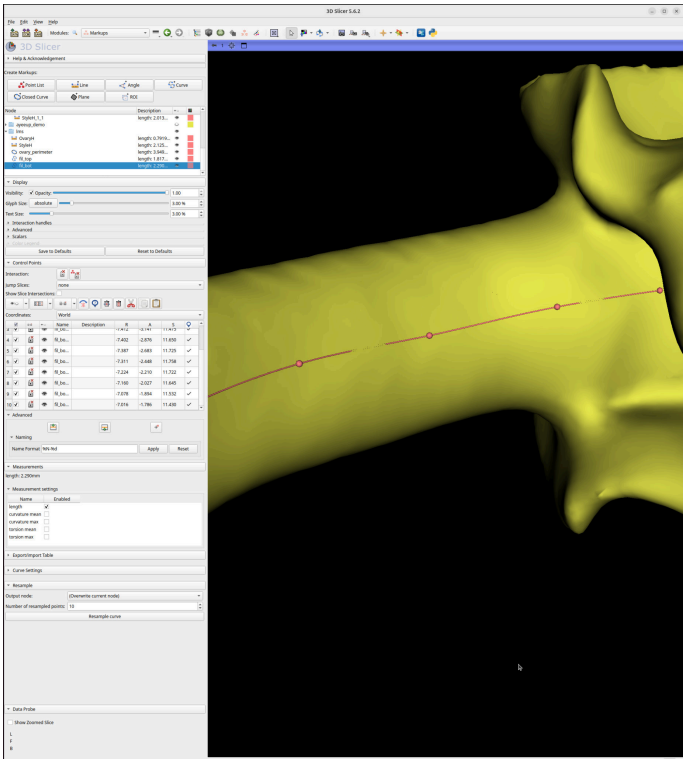
i. Click on “fil_bot”

b. Position the flower with a good view of the base of the filament

c. Place landmarks along the midpoint from the base of the filament to the base of the anther

i. Pause and toggle between the mouse and the landmark cursor to zoom in and out and move around to make sure points are along the midline





d. Scroll down to the “Resample” tab
    i. Select “fil_bot” as “Output Node”
    ii. Set the “Number of resampled points” to 15
    iii. Click “Resample curve”
    iv. Now all points are evenly spaced along the curve
e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

i. Make sure all landmark points have the closed lock icon next to them

10. Remember to save results periodically while landmarking!

a. Check that landmarks are being saved to the correct folder for your specimen

b. models_for_gmm/whole_flower/thecac/lms/[specimen_name]/lms/

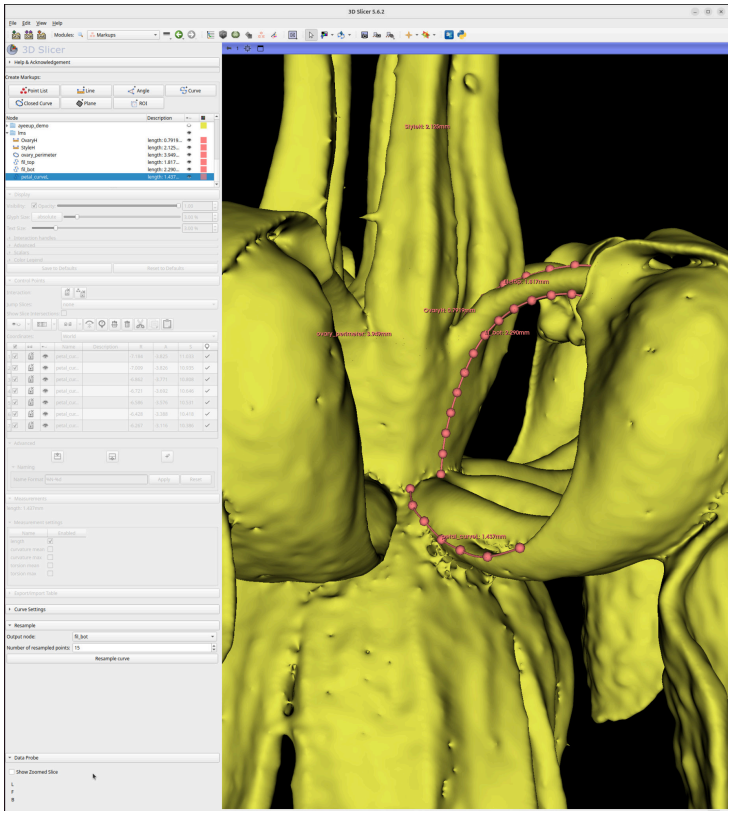
11. Click on the “Curve” tool under “Create Markups”

a. Rename it to “petal_curveL” (petal curve left)

i. Click on “petal_curveL”

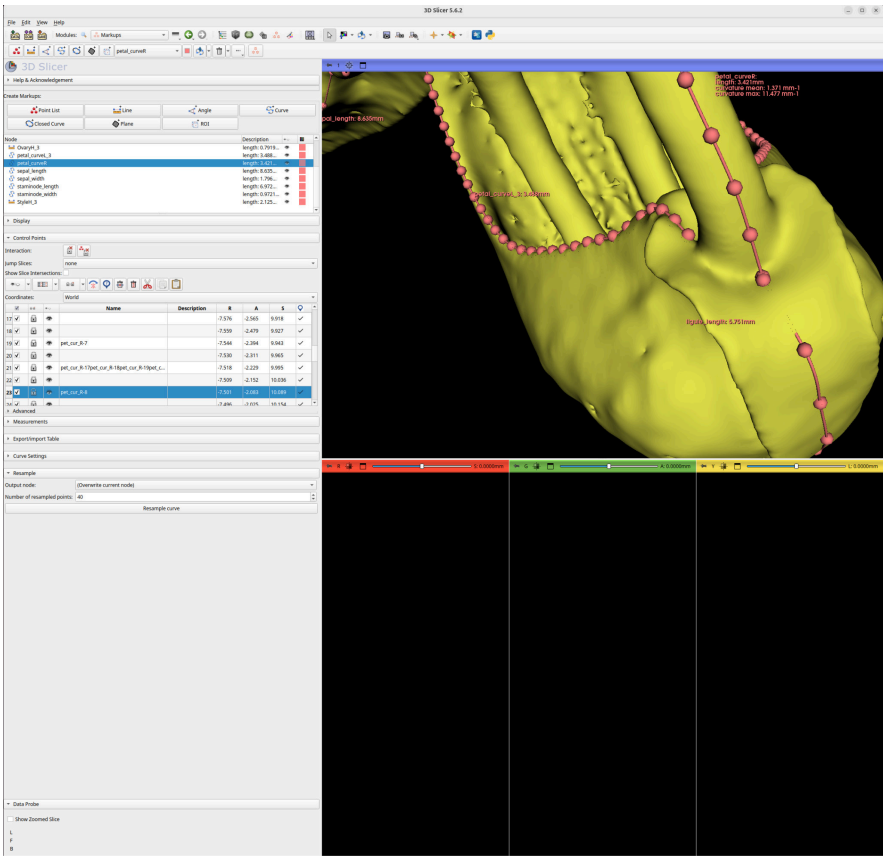
b. Position the flower with a good view of the base /area of insertion of the petal on the left side

c. Place landmarks along the center of the petal margin, all the way around the ear flap-like structure enclosing the anther.



i. Toggle between the landmarks cursor and the pointer to pause and rotate the flower for better access to landmark structures, as needed

ii. Stop at the end of the ear-like structure before the ligule starts

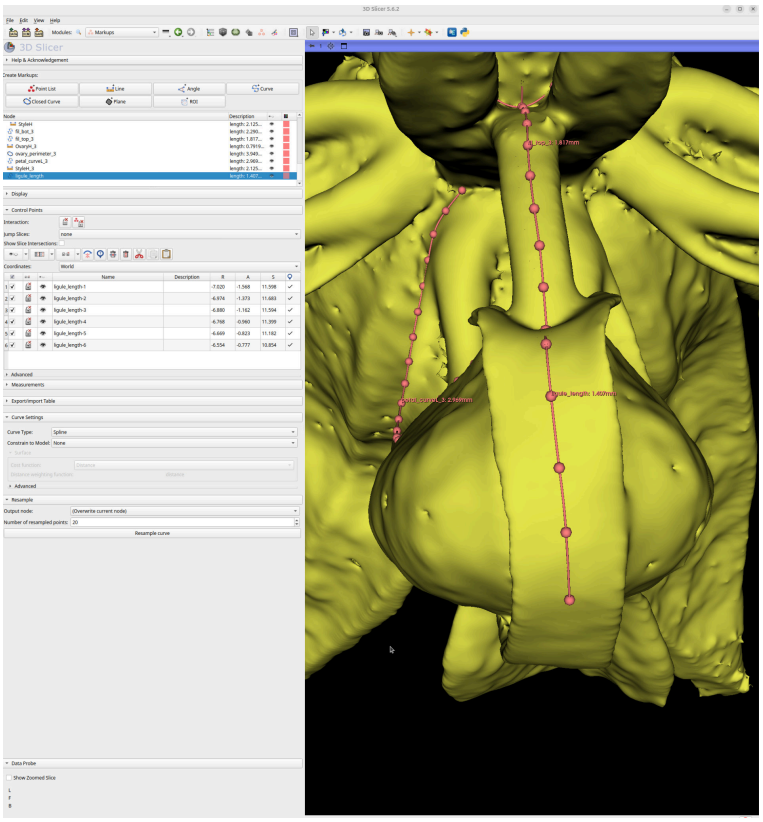


d. Scroll down to the “Resample” tab
    i. Select “petal_curveL” as “Output Node”
    ii. Set the “Number of resampled points” to 40
    iii. Click “Resample curve”
    iv. Now all points are evenly spaced along the curve
e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

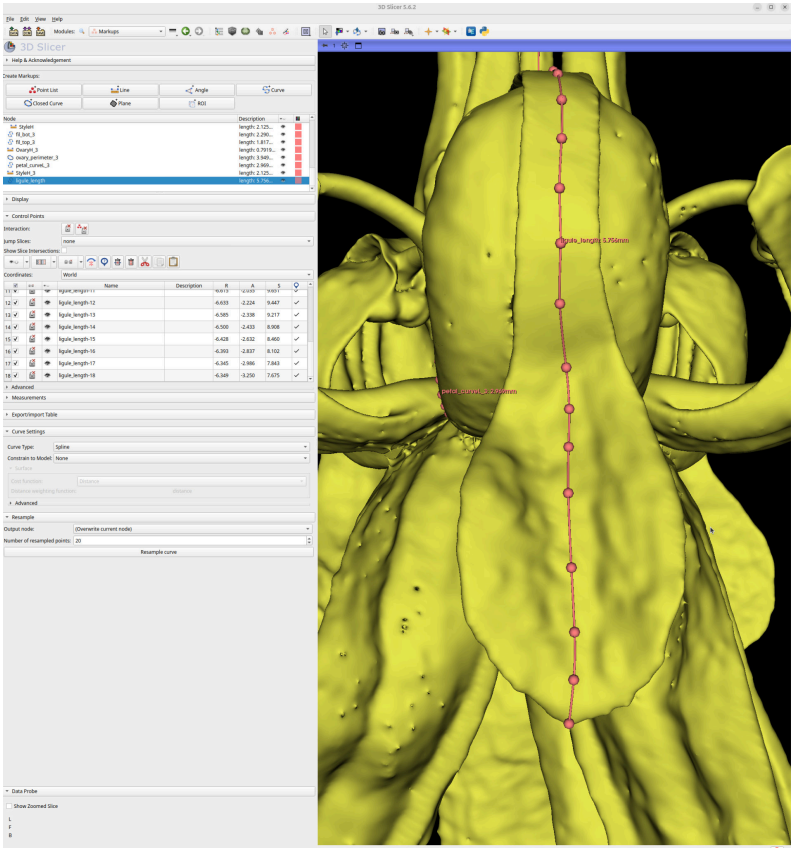
12. Repeat 9a-9e for “petal_curveR”

13. Click on the “Curve” tool under “Create Markups”

a. Rename it to “ligule_length”
    i. Click on “ligule_length”
b. Position the flower with a good view of the top of the petal/ area where the anther is sheathed
c. Place landmarks along the center of the ligule, from the top where the anther is inserted (proximal end) to the most distal end



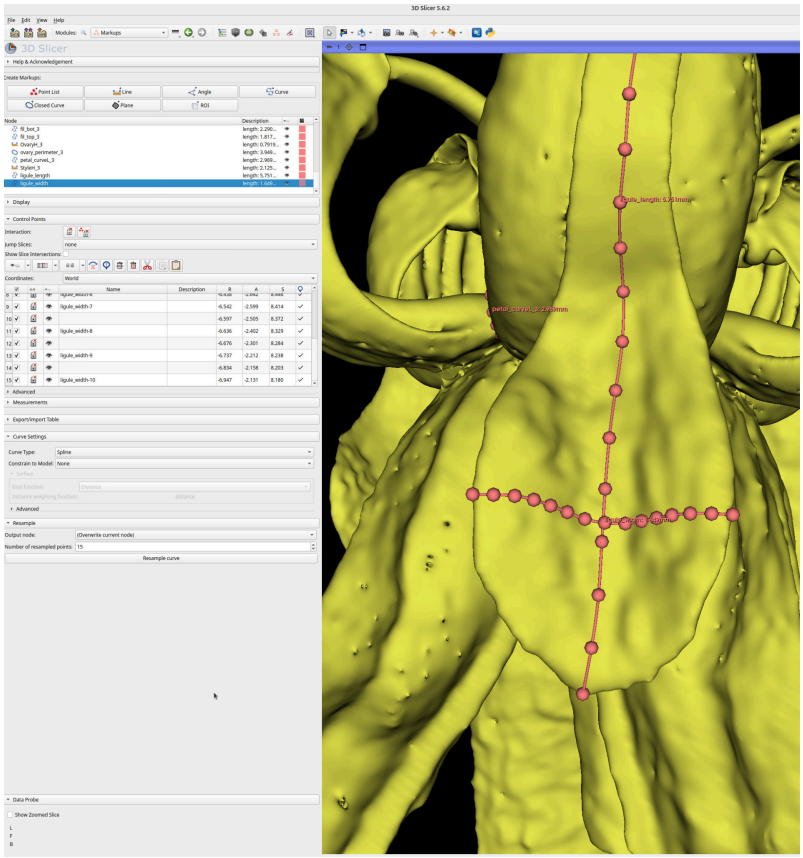
i. Toggle between the landmarks cursor and the pointer to pause and rotate the flower for better access to landmark structures, as needed



d. Scroll down to the “Resample” tab
    i. Select “ligule_length” as “Output Node”
    ii. Set the “Number of resampled points” to 20
    iii. Click “Resample curve”
    iv. Now all points are evenly spaced along the curve
e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

14. Click on the “Curve” tool under “Create Markups”

a. Rename it to “ligule_width”
    i. Click on “ligule_width”
b. Position the flower with a good view of the widest part of the ligule
c. Place landmarks along the midline of the widest part of the ligule, perpendicular to the line for ligule length

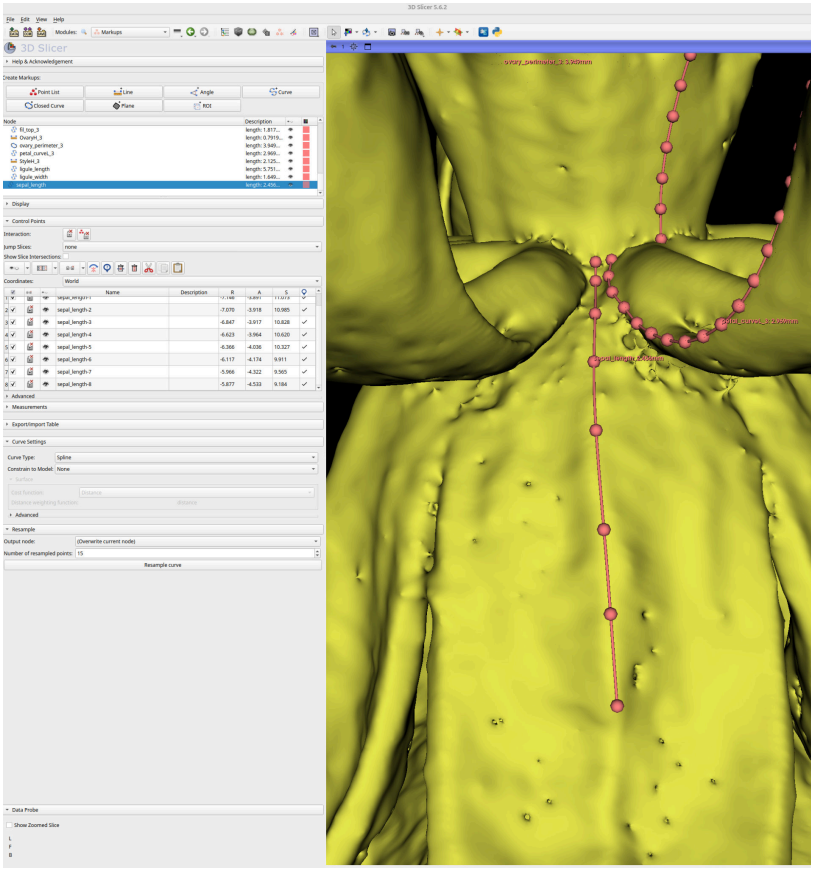


d. Scroll down to the “Resample” tab
    i. Select “ligule_width” as “Output Node”
    ii. Set the “Number of resampled points” to 15
    iii. Click “Resample curve”
    iv. Now all points are evenly spaced along the curve
e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

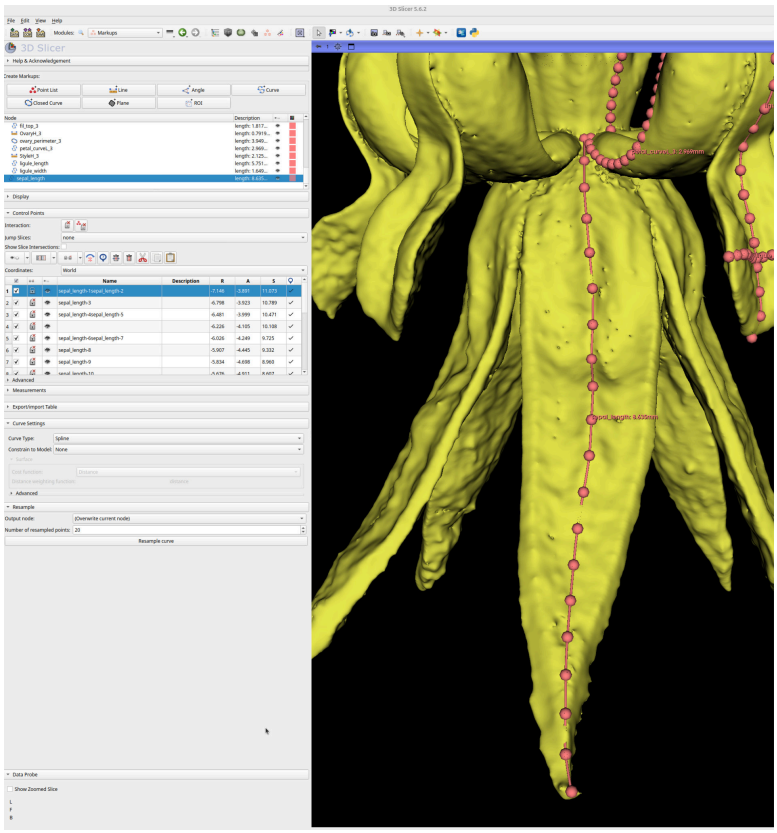
15. Remember to save results periodically while landmarking!

   a. **Check that landmarks are being saved to the correct folder for your specimen**
   b. **models_for_gmm/whole_flower/thecac/lms/[specimen_name]/lms/**

16. **Click on the “Curve” tool under “Create Markups”**
    a. **Rename it to “sepal_length”**
        i. Click on “sepal_length”
    b. Position the flower with a good view of the insertion of the sepal to the left of the chosen petal for landmarking
    c. Place landmarks along the center of the sepal, from the insertion on the staminal tube (proximal end) to the most distal end



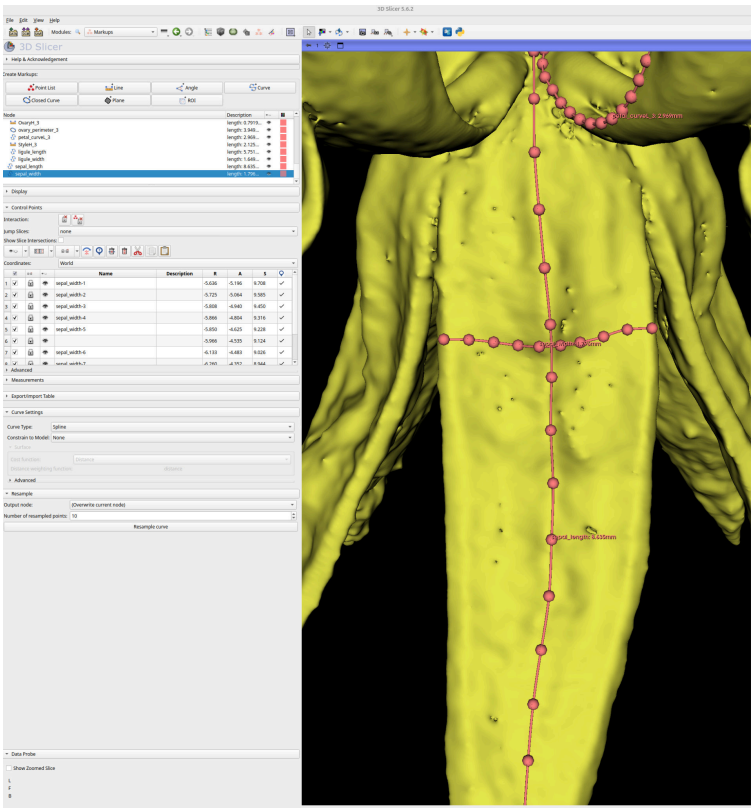
        i. Toggle between the landmarks cursor and the pointer to pause and rotate the flower for better access to landmark structures, as needed



d. Scroll down to the “Resample” tab
    i. Select “sepal_length” as “Output Node”
    ii. Set the “Number of resampled points” to 20
    iii. Click “Resample curve”
    iv. Now all points are evenly spaced along the curve
e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

17. Click on the “Curve” tool under “Create Markups”

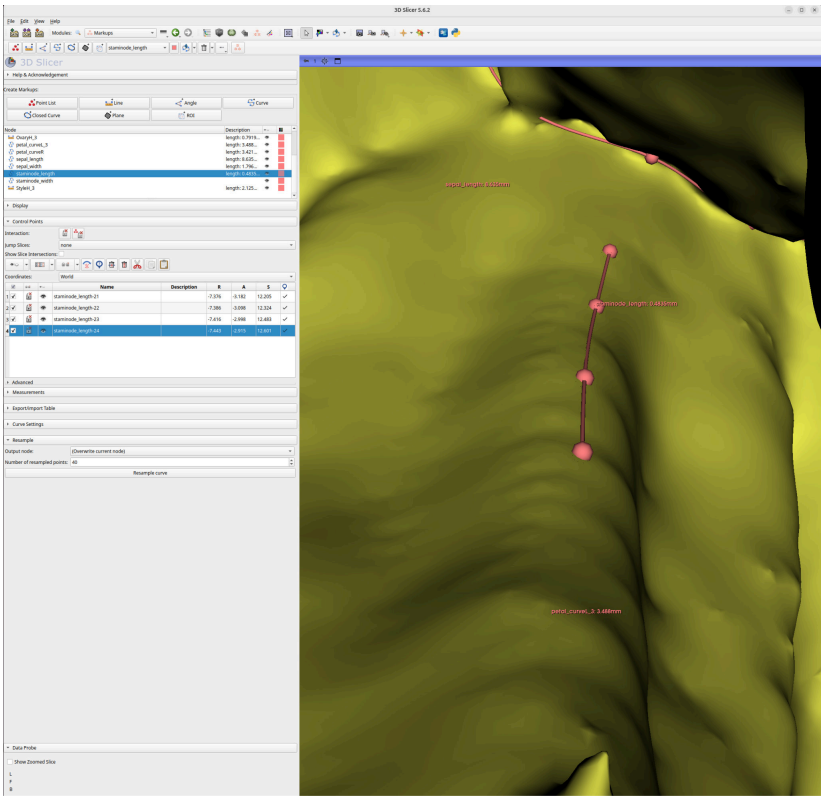
a. Rename it to “sepal_width”
    i. Click on “sepal_width”
b. Position the flower with a good view of the widest part of the sepal
c. Place landmarks along the midline of the widest part of the sepal, perpendicular to the line for sepal length
    i. Toggle between the landmarks cursor and the pointer to pause and rotate the flower for better access to landmark structures, as needed



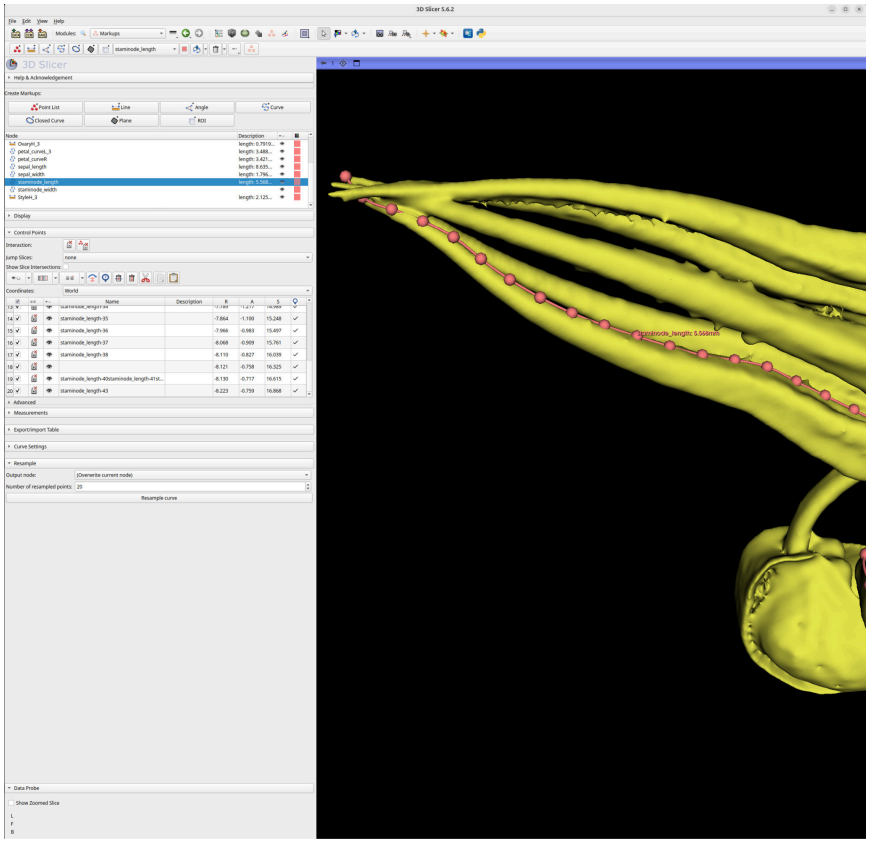
d. Scroll down to the “Resample” tab
    i. Select “sepal_width” as “Output Node”
    ii. Set the “Number of resampled points” to 10
    iii. Click “Resample curve”
    iv. Now all points are evenly spaced along the curve
e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

18. Click on the “Curve” tool under “Create Markups”

a. Rename it to “staminode_length”
    i. Click on “staminode_length”
        1. Copy the first point from “sepal_length” to “staminode_length”
        2. Rename it to “staminode_length-1”
        3. Click the lock icon next to “staminode_length-1”
b. Position the flower with a good view of the inner surface of the staminode to the left of the chosen petal for landmarking
c. Place landmarks along the center of the staminode, from the where it comes free from the staminal tube to the apex



i. Toggle between the landmarks cursor and the pointer to pause and rotate the flower for better access to landmark structures, as needed



d. Scroll down to the “Resample” tab
   i. Select “staminode_length” as “Output Node”

ii. Set the “Number of resampled points” to 20
iii. Click “Resample curve”
iv. Now all points are evenly spaced along the curve

e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

19. Click on the “Curve” tool under “Create Markups”

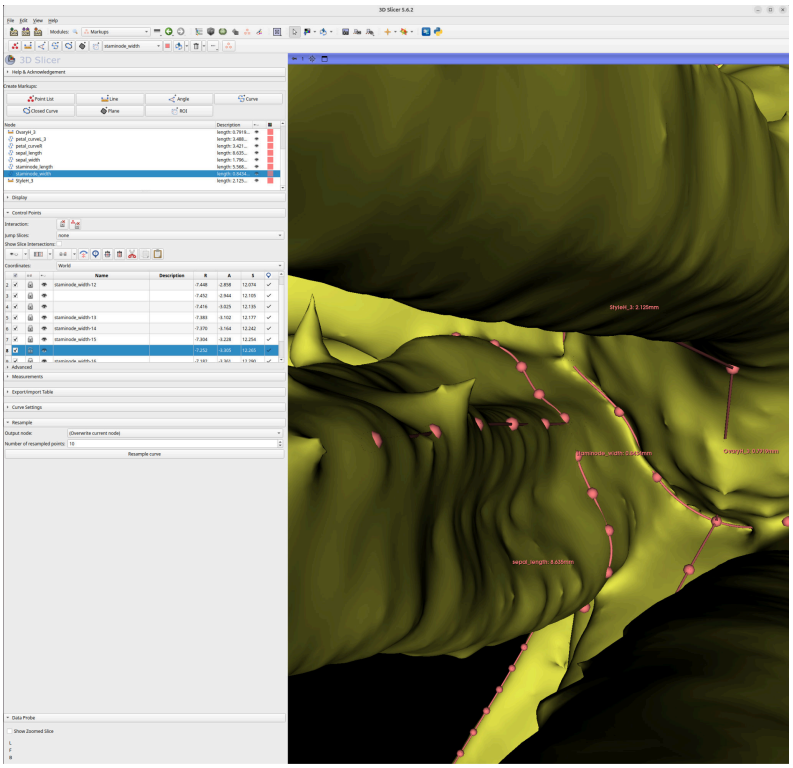
a. Rename it to “staminode_width”

i. Click on “staminode_width”

b. Position the flower with a good view of the widest part of the staminode

c. Place landmarks along the midline of the widest part of the staminode, perpendicular to the line for staminode length

i. Toggle between the landmarks cursor and the pointer to pause and rotate the flower for better access to landmark structures, as needed



d. Scroll down to the “Resample” tab

i. Select “staminode_width” as “Output Node”
ii. Set the “Number of resampled points” to 10
iii. Click “Resample curve”
iv. Now all points are evenly spaced along the curve

e. Click the lock icon above the lms coordinates table and select “Lock” to lock all coordinates and prevent accidental changes when clicking around the model

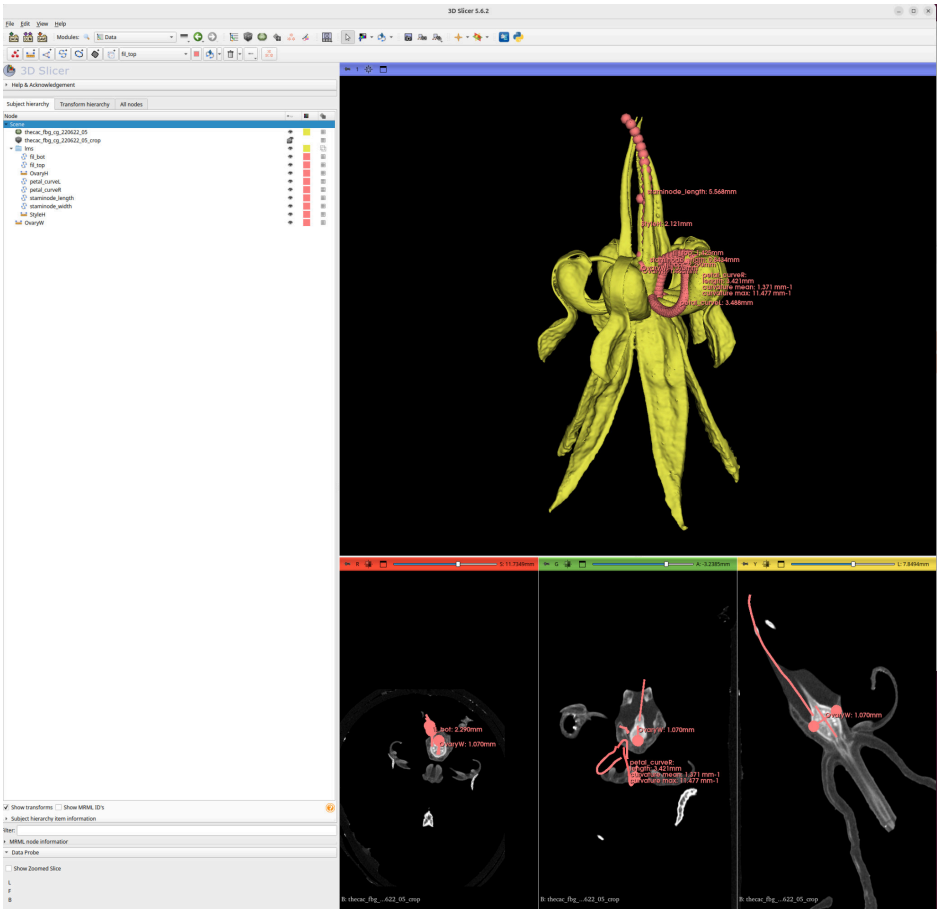
4. <u>Go to the “Data” Module</u>

a. Save your results

i. See recommended directory structure on Page 1

ii. Saving landmarks in the lms folder of your dataset of interest
    1. models_for_gmm/whole_flower/thecac/lms/[your dataset name]/lms/
    2. Move [your model].ply file to the “already lmed” (already landmarked) folder to help you keep track of which datasets are remaining
        a. models_for_gmm/whole_flower/thecac/already_lmed/

b. Marvel at your beautiful landmarks



c. Close 3D Slicer to clear the cache and reduce risk of software crashes from too much data left in the memory
d. Repeat